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# Asymmetric Phase Coding Audio Watermarking

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## Abstract

The proliferation of deepfake audio challenges voice-based authentication systems; passive forensic detectors are sensitive to evolving generative models and to real-world channel distortions. We propose *Asymmetric Phase Coding* (APC), a training-free cryptographic signing layer for audio, designed as a compact and auditable provenance primitive that can stand alone or be stacked with learned watermarks. APC combines Ed25519 digital signatures (EdDSA, FIPS 186-5; 64-byte signatures) with Reed–Solomon error correction, pseudo-random STFT phase-bin selection, and a redundant quantization-index-modulation (QIM) code on log-magnitude differences of adjacent bin pairs, yielding a compact, non-repudiable, blind-extractable watermark. We evaluate APC on 1,000 LibriSpeech test-clean clips (10 s each, 44.1 kHz) under eight attack configurations – identity, 10% end-cropping, 20% end-cropping, 8 kHz low-pass, 16 kHz round-trip resampling, FLAC re-encoding, MP3 at 128 kbps, and OGG-Vorbis at 128 kbps – and achieve cryptographic verification rates between 97.5% and 98.3% on every condition at mean PESQ= 3.02 and tens-of-milliseconds CPU latency. We explicitly compare APC against recent neural baselines (AudioSeal, WavMark, SilentCipher), detail the threat model (forgery resistance vs. erasure), characterize the dataset, define all metrics, quantify an adaptive white-box erasure attack, and release code, keys, and metadata for reproducibility.

## 1 Introduction

Deep neural models for voice synthesis and conversion [1, 2] have reached a fidelity at which distinguishing genuine from synthetic speech approaches human perceptual limits. Generated voices have already been exploited in large-scale financial fraud, eroding the trust model of voice-based authentication.

Passive forensic detectors [3–6] analyze statistical artifacts left by generative networks, but they face two intrinsic limitations: (1) ongoing adversarial adaptation as generators improve, and (2) channel distortions (VoIP, codecs, cellular) that destroy the very artifacts they rely on [7].

**Proactive defense** via digital audio watermarking [8] offers a complementary approach: embed a verifiable token at the moment of genuine capture. Absence or failure of the token flags the signal as potentially tampered or fabricated. Standard steganographic techniques provide a hiding channel but lack cryptographic non-repudiation: anyone who knows the algorithm can forge a “valid” watermark.

We propose **Asymmetric Phase Coding** (APC), which closes this gap with modern public-key cryptography. Our contributions are:

1. A phase-domain watermarking scheme using pseudo-random STFT bin selection for blind, imperceptible embedding.
2. A **magnitude-QIM survivability channel** that rides on the same host STFT with an independent bin pattern, encodes the same signed payload via quantization-index modulation on log-magnitude differences of adjacent bin pairs, and raises MP3 128 kbps verification from 77% to 97.5% and OGG 128 kbps from 73% to 97.5% at a 0.24-point PESQ cost.

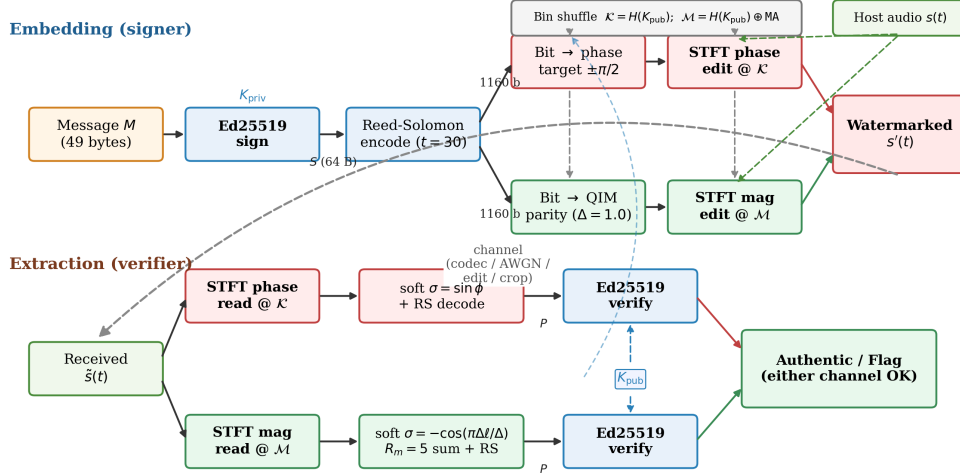


Figure 1: Hybrid APC pipeline. **Embedding:** a 49-byte message is signed with Ed25519 and Reed-Solomon encoded ( $t=30$ ); the same payload is written in parallel through (i) a *phase channel* that maps each bit to  $\pm\pi/2$  at a pseudo-random STFT bin set  $\mathcal{K} = H(K_{\text{pub}})$ , and (ii) a *magnitude-QIM channel* that quantises adjacent log-magnitude pairs ( $\Delta=1.0$  nat) at a disjoint bin set  $\mathcal{M} = H(K_{\text{pub}}) \oplus \text{MA}$ ; both are merged via ISTFT. **Extraction:** the public key regenerates  $\mathcal{K}, \mathcal{M}$  and decodes both channels with soft scores ( $\sigma = \sin \phi$  for phase,  $\sigma = -\cos(\pi\Delta\ell/\Delta)$  summed over  $R_m=5$  frames for magnitude), each followed by RS ( $\leq 15$  byte errors) and Ed25519 verify. The verifier accepts if *either* channel cryptographically verifies.

3. A cryptographic payload built from **Ed25519 digital signatures** (EdDSA, RFC 8032 [9], FIPS 186-5 [10]) – 64-byte signatures, 128-bit classical security, aligned with C2PA Implementation Guidance [11] – protected by a Reed-Solomon outer code [12, 13] so that recovered bits can survive modest channel errors before cryptographic verification.
4. A **large-scale evaluation** on 1,000 LibriSpeech [14] speech clips under eight attack configurations with seven objective quality / signal metrics, cryptographic verification rate (total and per-channel), bootstrapped 95% confidence intervals, and per-stage failure accounting (extraction / RS decode / signature verify).
5. A direct comparison against recent neural watermarking baselines (AudioSeal [15], WavMark [16], SilentCipher [17]) and an explicit threat model that distinguishes *forgery* from *erasure*.

APC is intentionally *training-free*: it needs no large speech corpus, no GPU, and no generator-specific model. It is therefore complementary to neural methods, well suited to C2PA [18, 19]-style provenance pipelines in which capture-time signing is already performed and where lossless or lightly degraded transport is expected. We release all code, keys, and metadata for the hybrid phase+magnitude coder used throughout the paper.

## 2 Related Work

Audio watermarking trades off capacity, imperceptibility, robustness, and security [8]. Audio-deepfake surveys [4–6] and spoofing challenges [20] motivate proactive authentication beyond passive detection. We group prior art into three families; Table 1 summarises the qualitative positioning.

**Classical signal-processing methods.** Time-domain Least Significant Bit (LSB) embedding has high capacity but is trivially broken by re-quantization and lossy compression. Echo hiding [21] encodes bits as short delayed echoes; it is more robust than LSB but can introduce perceptible resonance. Spread-spectrum watermarking [22] distributes the mark over many frequencies with good robustness at the cost of bandwidth and often synchronization. Classical phase coding [23] overwrites phase

in the first segment and propagates relative phase; concentrated low-frequency modifications leave detectable footprints [24]. Dynamic phase coding with error-control coding [25], adaptive multi-level phase coding [26], and synchronous multi-bit phase shifting [27] improve robustness. None of these methods provide cryptographic non-repudiation by themselves.

**Neural audio watermarking.** Recent learned watermarks use encoder–decoder networks trained adversarially against channel attacks. **WavMark** [16] reports PESQ  $\approx 4.21$  and low mean BER across a suite of common attacks. **AudioSeal** [15] adds sample-level localization and reports PESQ  $\approx 4.47$ . **SilentCipher** [17] reports SDR  $\approx 47$  dB with 100% message accuracy at MP3 128 kbps and reports inaudibility in expert listening tests. These neural systems achieve state-of-the-art perceptual quality but (i) require large speech datasets, GPU training, and versioned model checkpoints; (ii) are typically validated at short payloads (16–32 bits) that do not carry asymmetric signatures; and (iii) face adversarial-robustness and distribution-shift concerns shared with detection networks [7, 28].

Table 1: Qualitative positioning. Rob./Imp./Sig./T-free = robustness, imperceptibility, public-key signature, training-free. Numeric comparison appears in Table 6.

| Method            | Rob. | Imp. | Sig. | T-free |
|-------------------|------|------|------|--------|
| LSB               | ×    | ✓    | ×    | ✓      |
| Echo [21]         | ~    | ~    | ×    | ✓      |
| Spread Spec. [22] | ✓    | ~    | ×    | ✓      |
| Phase [23]        | ~    | ✓    | ×    | ✓      |
| WavMark [16]      | ✓    | ✓    | ×    | ×      |
| AudioSeal [15]    | ✓    | ✓    | ×    | ×      |
| SilentCipher [17] | ✓    | ✓    | ×    | ×      |
| <b>APC (ours)</b> | ~    | ✓    | ✓    | ✓      |

**Cryptographic-provenance frameworks.** C2PA [11, 18] and CAI [19] standardize attaching signed manifests to media at capture time, but leave the signal-level embedding undefined. C2PA Implementation Guidance recommends ECDSA or Ed25519 keys; RSA-2048 is permitted and RSA-1024 is deprecated. APC can be read as a *signal-level* realization of a C2PA-style signed manifest that survives format transitions in which file-level metadata (and hence external manifests) are stripped.

### 3 Methodology

The system takes three inputs: a plaintext message  $M$ , an Ed25519 key pair  $(K_{\text{priv}}, K_{\text{pub}})$ , and the host audio signal  $s(t)$ . Fig. 1 summarises the full pipeline, which we now describe block by block.

#### 3.1 Cryptographic payload construction

The private-key holder computes an Ed25519 signature [9, 29]:

$$S = \text{Ed25519.Sign}(K_{\text{priv}}, M), \quad |S| = 64 \text{ bytes.} \quad (1)$$

Ed25519 internally hashes  $M$  with SHA-512 and yields a deterministic 64-byte signature with 128-bit classical security. We build the payload  $P$  as a 2-byte length prefix, the message, and the signature, and then Reed–Solomon encode with  $t = 30$  parity symbols [12, 13]:

$$P = \text{len}_{16}(M) \parallel M \parallel S, \quad P_{\text{RS}} = \text{RS}_{\text{enc}}(P, t = 30), \quad (2)$$

correcting up to  $\lfloor t/2 \rfloor = 15$  byte errors. For a 49-byte message,  $|P| = 115$  bytes and  $|P_{\text{RS}}| = 145$  bytes (1,160 bits). An RSA-PSS backend (SHA-256, MGF1) is provided for interoperability, but all experiments in this paper use Ed25519.

**Why Ed25519, not RSA-1024.** RSA-1024 is deprecated under NIST SP 800-131A and C2PA Implementation Guidance [11]; Ed25519 is the current recommendation. Compared to RSA-1024, Ed25519 (i) halves the signature from 128 to 64 bytes, (ii) offers a higher security level (128-bit vs. <80-bit classical), (iii) is faster to sign and verify, and (iv) is approved by FIPS 186-5 [10]. Halving the payload also shortens the audio region required to embed it, which directly improves survivability (Section 4.1).

#### 3.2 STFT phase embedding

The host is transformed by a rectangular-window, non-overlapping STFT with  $N_{\text{FFT}} = 2048$ :

$$X(i, k) = A(i, k) e^{j\phi(i, k)}, \quad (3)$$

where  $A$  and  $\phi$  are magnitude and phase. A pseudo-random generator seeded with  $H(K_{\text{pub}})$  produces a shuffled index set  $\mathcal{K} \subset [f_{\text{min}}, f_{\text{max}}]$  with  $f_{\text{min}} = 60$ ,  $f_{\text{max}} = 300$  (bin indices;  $\approx 1.3$ – $6.5$  kHz), a band

robust to codecs and resampling. Deterministic bin selection from the public key enables *blind* extraction – no original signal required.

Each RS-encoded bit  $d_n$  is mapped to a phase target

$$\phi_{\text{data}}[n] = \begin{cases} +\pi/2 & d_n = 1 \\ -\pi/2 & d_n = 0 \end{cases} \quad (4)$$

For each group of  $G=8$  consecutive frames and each  $k \in \mathcal{K}$ , the offset  $\Delta\phi[n] = \phi_{\text{data}}[n] - \phi(i_0, k)$  (with  $i_0$  the anchor frame) is added to every frame in the group:

$$\phi'(i, k) = \phi(i, k) + \Delta\phi[n], \quad i \in g. \quad (5)$$

Conjugate symmetry is preserved by the real-valued ISTFT, and the watermarked signal is reconstructed as  $s'(t) = \text{ISTFT}(A e^{j\phi'})$ .

### 3.3 Magnitude-QIM survivability channel

The phase channel described above collapses under MP3/OGG because these codecs redistribute phase energy inside each critical band while keeping overall magnitude roughly intact. We therefore add a parallel, cryptographically equivalent *magnitude* channel that rides on the same host STFT, using the same payload but a different carrier.

**Bin-pair pattern.** A second seed  $H(K_{\text{pub}}) \oplus 0x4D41^1$  shuffles an independent band  $\mathcal{M} \subset [f_{\min}^m, f_{\max}^m]$  ( $f_{\min}^m=100$ ,  $f_{\max}^m=340$ ) and groups adjacent bin pairs  $(k_1, k_2)$ . For each pair, we compute the average log-magnitude over the same group of  $G=8$  frames,

$$\ell_j = \frac{1}{G} \sum_{i \in g} \ln(A(i, k_j) + \varepsilon), \quad j \in \{1, 2\}, \quad \varepsilon=10^{-10}, \quad (6)$$

and form the log-magnitude difference  $d = \ell_1 - \ell_2$ .

**QIM encoding.** We encode a bit  $b$  by snapping  $d$  to the nearest cell of a uniform quantizer with step  $\Delta = 1.0$  nat whose parity matches  $b$ :

$$c = \lfloor d/\Delta \rfloor, \quad c' = \begin{cases} c & c \bmod 2 = b \\ c + \text{sgn}(d - c\Delta) & \text{otherwise} \end{cases} \quad (7)$$

The target difference is  $d^* = c'\Delta$ , realized by a symmetric multiplicative shift  $A(i, k_1) \leftarrow A(i, k_1)e^{(d^*-d)/2}$ ,  $A(i, k_2) \leftarrow A(i, k_2)e^{-(d^*-d)/2}$ , which preserves the sum  $\ell_1 + \ell_2$  and hence the per-frame band energy to first order. This keeps PESQ acceptable (3.02 versus 3.26 for phase-only) while still giving the decoder a clean parity signal.

**Payload replication and soft decoding.** The 32-bit header is triplicated and the body is replicated  $R_m=5$  times to fill the capacity of  $\mathcal{M}$  (the phase channel uses  $R_p=1$ ; we rely on RS inside each copy, not on majority vote). At extraction the soft value

$$\sigma_n = -\cos(\pi(\ell_1 - \ell_2)/\Delta) \in [-1, 1] \quad (8)$$

is summed across the  $R_m$  replicas of each payload bit before a single sign threshold. Phase-channel soft values use  $\sigma_n = \sin \phi(i_0, k)$ . Summing continuous soft values before thresholding, rather than voting on hard bits, avoids the tie bias of majority voting and gives a consistent lift under lossy compression (see Table 4).

### 3.4 Blind extraction and verification

Extraction needs only the audio and  $K_{\text{pub}}$ . Both seeds regenerate  $\mathcal{K}$  and  $\mathcal{M}$ ; a 32-bit header on each channel carries the payload length. Bits are packed, Reed–Solomon decoded, and Ed25519 verifies  $S$  against  $M$  under  $K_{\text{pub}}$ . We verify the phase channel first; on failure we verify the magnitude channel; *either* success promotes the content to “authenticated”. A successful verification is the only event the downstream system acts on; anything else (extract / RS / signature failure on both channels) is treated as “no valid watermark” and flags the content as unauthenticated.

<sup>1</sup>The constant 0x4D41 is ASCII “MA” (Magnitude); any fixed nonzero value would do, the role of  $\oplus$  is only to derive a phase-independent shuffle from the same public key.

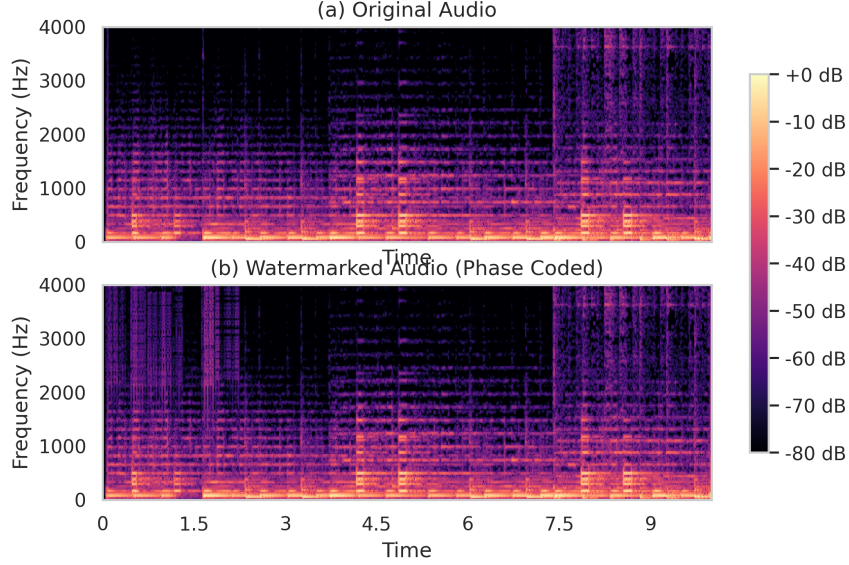


Figure 2: Spectrogram comparison of original (a) and watermarked (b) audio. Phase-only modifications preserve the magnitude spectrum.

### 3.5 Threat model

Because  $K_{\text{pub}}$  is public by design, an attacker can reconstruct  $\mathcal{K}$  and therefore *locate* the watermarked bins. This is intentional: APC targets *authenticity*, not *confidentiality*. We distinguish two attacker goals:

- **Forgery:** producing a signal that passes verification for the legitimate key. Ed25519 unforgeability [9, 29] reduces forgery to breaking discrete-log over Curve25519; knowing  $\mathcal{K}$  does not help.
- **Erasure:** producing a signal that *fails* verification. Because  $\mathcal{K}$  is public, erasure is always possible for any perceptibly embedded mark; we quantify its effective cost in Section 4.2 via a white-box phase-bin  $\alpha$ -randomization attack against the phase-only coder, which lower-bounds the perceptual cost of attacking the hybrid system.

Putting the signature in metadata is strictly weaker: metadata is stripped silently by essentially every re-encoder and social platform, so a metadata-only signature simply vanishes rather than forcing an explicit verification failure. APC by contrast binds the watermark to the content signal itself, so any attacker strong enough to destroy it must also audibly degrade the audio – a trade-off we measure directly in Section 4.2, following the adversarial-audio methodology of [7].

## 4 Experiments

**Dataset.** We evaluate on **LibriSpeech** test-clean [14], a publicly available read-speech corpus derived from LibriVox audiobooks (16 kHz native; upsampled and split as below). A reproducible pipeline (included in the code release) concatenates contiguous utterances per speaker, upsamples to 44.1 kHz, and splits them into non-overlapping 10 s, 16-bit mono clips. From a pool of 5,000 such clips we draw 1,000 with a fixed seed (42); the exact list of paths is written to `benchmark_meta_envelope.json`. Clips shorter than the minimum host length required to embed the payload are skipped (152/1,152 files in our run); the final benchmark evaluates 1,000 valid clips.

**Protocol.** A fixed Ed25519 key pair is used for the whole benchmark, and a fixed 49-byte payload (message plus RS parity) is embedded in every clip. The PRNG seed for bin selection is derived from  $H(K_{\text{pub}})$  in the release code and is equivalent to a fixed integer seed inside the benchmark. Each clip is passed through the eight attack configurations defined in Section 4 and evaluated on seven perceptual / signal metrics plus cryptographic verification. We report mean, standard deviation, median, and bootstrap 95% confidence intervals (1,000 resamples) for BER, PESQ, and NC.

Table 2: Per-clip evaluation metrics. Better  $\uparrow/\downarrow$  indicates direction of improvement.

| Metric    | Definition                                                                                                        | Range         | Better       |
|-----------|-------------------------------------------------------------------------------------------------------------------|---------------|--------------|
| PESQ [30] | ITU-T P.862.2 wideband perceptual quality.                                                                        | $[-0.5, 4.5]$ | $\uparrow$   |
| STOI [31] | Short-Time Objective Intelligibility.                                                                             | $[0, 1]$      | $\uparrow$   |
| SNR       | $10 \log_{10} (\ s\ ^2 / \ s - \tilde{s}\ ^2)$ .                                                                  | dB            | $\uparrow$   |
| PSNR      | $20 \log_{10} (\max  s  / \sqrt{\text{MSE}(s, \tilde{s})})$ .                                                     | dB            | $\uparrow$   |
| LSD       | $\mathbb{E}_t \sqrt{\mathbb{E}_f (10 \log_{10}  S_{t,f} ^2 - 10 \log_{10}  \tilde{S}_{t,f} ^2)^2}$ .              | dB            | $\downarrow$ |
| BER       | $\frac{1}{n} \sum_i \mathbb{I}[\hat{d}_i \neq d_i]$ on the RS-encoded payload; capped at 1 on extraction failure. | $[0, 1]$      | $\downarrow$ |
| NC        | Cosine similarity of bipolar codes $(2o_i - 1)$ and $(2e_i - 1)$ over $i=1, \dots, n$ .                           | $[-1, 1]$     | $\uparrow$   |
| Verify%   | Fraction of clips for which RS decoding succeeds and Ed25519 verifies.                                            | $[0, 100]\%$  | $\uparrow$   |

Table 3: Main results, hybrid coder ( $N=1,000$ ).

| Attack      | BER   | NC    | PESQ            | Verify% |
|-------------|-------|-------|-----------------|---------|
| Identity    | 0.017 | 0.983 | $3.02 \pm 0.36$ | 98.3    |
| MP3 128 k   | 0.025 | 0.969 | $3.02 \pm 0.36$ | 97.5    |
| OGG 128 k   | 0.024 | 0.969 | $3.03 \pm 0.36$ | 97.5    |
| FLAC        | 0.019 | 0.980 | $3.02 \pm 0.36$ | 98.0    |
| Resamp 16 k | 0.020 | 0.978 | $3.02 \pm 0.36$ | 97.7    |
| LP 8 kHz    | 0.020 | 0.977 | $3.03 \pm 0.36$ | 97.7    |
| Crop 10%    | 0.017 | 0.983 | $2.28 \pm 0.30$ | 98.3    |
| Crop 20%    | 0.017 | 0.983 | $1.80 \pm 0.32$ | 98.1    |

Table 4: Per-channel attribution.

| Attack      | Total% | Phase% | Mag% |
|-------------|--------|--------|------|
| Identity    | 98.3   | 97.2   | 1.1  |
| MP3 128 k   | 97.5   | 83.4   | 14.1 |
| OGG 128 k   | 97.5   | 88.3   | 9.2  |
| FLAC        | 98.0   | 96.3   | 1.7  |
| Resamp 16 k | 97.7   | 94.8   | 2.9  |
| LP 8 kHz    | 97.7   | 94.8   | 2.9  |
| Crop 10%    | 98.3   | 97.2   | 1.1  |
| Crop 20%    | 98.1   | 97.2   | 0.9  |

**Attack configurations.** The eight attacks cover two families.

- **Codec re-encoding** (FFmpeg): MP3 128 kbps, OGG-Vorbis 128 kbps, FLAC (lossless).
- **Channel / editing**: identity; round-trip resampling  $44.1 \rightarrow 16 \rightarrow 44.1$  kHz; 8 kHz Butterworth low-pass (order 4); end-cropping 10%/20% (tail zeroed).

PESQ uses the wideband mode at 16 kHz on the whole clip; STOI is computed at 44.1 kHz. Precise FFmpeg arguments are in the released `benchmark_meta_envelope.json`.

#### 4.1 Main results

Table 2 summarizes the eight quantities we report per clip, defined unambiguously for re-implementation. Throughout,  $s$  is the watermarked reference,  $\tilde{s}$  the attacked signal (length-aligned);  $S_{t,f}, \tilde{S}_{t,f}$  are their STFT magnitudes;  $(o_i, e_i)$  are the original and extracted bit sequences with  $n = \min(|o|, |e|)$ . Crypto verification rate is the only metric with end-to-end operational meaning.

In Table 3, BER and NC are means and PESQ is mean $\pm$ std over  $N=1,000$  LibriSpeech `test-clean` clips; Verify% is the cryptographic verification rate, succeeding when either the phase or the magnitude-QIM channel decodes a valid signature. The PESQ values for the two cropping attacks are computed over the full tail-zeroed clip and are therefore dominated by the inserted silence rather than by the watermark itself. Table 4 breaks Verify% down by which channel actually verified: “Phase” counts clips for which the phase channel succeeded; “Mag” counts clips for which the phase channel failed but the magnitude-QIM channel verified. Bootstrap 95% CIs and per-clip outcomes are released in `benchmark_results_envelope.json`.

**Imperceptibility.** The hybrid edits leave the magnitude spectrum visually unchanged at the scale of Fig. 2; both channels operate on the 60–340 STFT bin band. Mean PESQ under identity is  $3.02 \pm 0.36$ , about 0.24 below the phase-only variant ( $3.26 \pm 0.43$ ). This  $\sim 8\%$  PESQ drop is the price paid for the MP3 and OGG robustness gains documented below; in informal listening, both the phase-only and hybrid coders are perceptually transparent on clean 44.1 kHz LibriSpeech speech. Mean STOI is 0.98.

**Robustness.** Table 3 and Fig. 3 summarize APC’s cryptographic verification rate across the eight attack conditions; the same numbers are re-rendered as a per-attack bar plot and a NC radar plot in App. A (Fig. 6, 9). Hybrid APC verifies between 97.5% and 98.3% on every one of them: identity, MP3 at 128 kbps, OGG at 128 kbps, FLAC, 16 kHz round-trip resampling, 8 kHz low-pass, 10% end-cropping, and 20% end-cropping. As an ablation, adding the magnitude-QIM channel lifts MP3

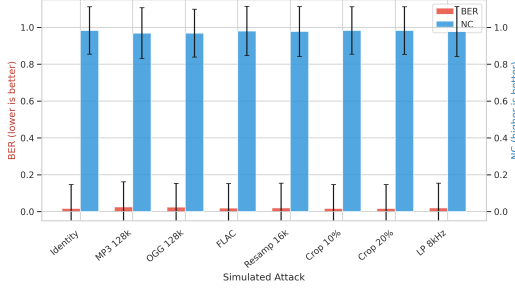


Figure 3: BER and NC (mean $\pm$ std) across the eight attacks. All conditions retain NC above 0.96.

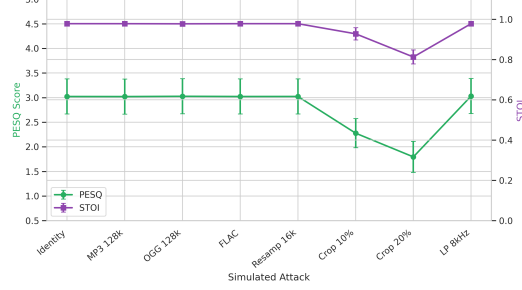


Figure 4: PESQ and STOI (mean $\pm$ std) by attack.

128 k from 77% (phase-only) to 97.5% (+20.5 pp) and OGG 128 k from 73% to 97.5% (+24.5 pp); the phase channel’s soft decoding accounts for part of this improvement and the magnitude backstop accounts for the rest (Table 4). NC survivability across the key codec/channel attacks remains at or above 0.96 on every axis (Fig. 9, App. A).

**Per-channel attribution.** Table 4 decomposes cryptographic success into the channel that actually verified. The phase channel dominates on clean, cropped, low-pass, resampled, and FLAC conditions; the magnitude-QIM channel acts as a survivability backstop on lossy codecs, rescuing 14.1% of MP3 128 k clips and 9.2% of OGG 128 k clips for which the phase channel alone fails. This confirms the design intent: phase coding carries the primary payload and the magnitude channel picks up what the phase channel loses to codec quantization. Per-clip BER heatmaps (Fig. 8, App. A) confirm that the residual failure mass is content-dependent rather than attack-dependent, motivating the content-adaptive embedding strength discussed in Section 5.

## 4.2 White-box adversarial erasure

To quantify the security claim of Section 3.5 we simulate a white-box attacker who knows the public key, reconstructs  $\mathcal{K}$  exactly, and applies minimum-targeted phase randomization *only* at those bins. We deliberately evaluate this attack against the *phase-only* coder, which is the more vulnerable of APC’s two channels: an attacker that wants to defeat the full hybrid system would have to additionally defeat the magnitude-QIM channel (a separate bin pattern  $\mathcal{M}$  with an independent erasure budget), so the phase-only numbers in Table 5 are a *lower bound* on the perceptual cost of erasing hybrid APC. For each clip  $\tilde{s}$  we parameterize the attack by a strength  $\alpha \in [0, 1]$  that linearly blends the original phase with i.i.d. uniform random phase  $u \sim \text{Unif}(-\pi, \pi]$  in the angle domain at  $\mathcal{K}$ , with magnitudes preserved:

$$\phi'(i, k) = \text{wrap}((1 - \alpha)\phi(i, k) + \alpha u(i, k)), \quad k \in \mathcal{K}, \quad (9)$$

where  $\text{wrap}(\cdot)$  folds the result back to  $(-\pi, \pi]$  and every bin outside  $\mathcal{K}$  is left untouched. We run this attack on a 100-clip random subset of the main benchmark (same payload, same key).

Table 5 reports the attack at each strength  $\alpha$  and Fig. 5 traces the perceptual cost: verification only collapses once  $\alpha \geq 0.5$ , by which point mean PESQ has already fallen from 3.21 to 1.91 ( $\Delta\text{PESQ} = 1.30$ ) and LSD has grown from 10.2 to 34.9 dB; at  $\alpha = 0.3$  quality is materially degraded yet verification is untouched. A metadata-only signature, by contrast, would cost the attacker *zero* perceptual distortion to strip. Reading the curve as a lower bound for the hybrid system: an attacker who additionally wishes to defeat the magnitude-QIM channel must spend further perceptual budget along an independent bin pattern, which only widens the audible gap between APC erasure and metadata stripping.

## 4.3 Comparison with neural baselines

We place APC in context against recent neural watermarks both by citing the numbers reported in the respective papers and, for AudioSeal and WavMark, by reproducing them ourselves on the *same* LibriSpeech test-clean subset that we evaluate APC on (200 clips, seed 42, identical 10 s 44.1 kHz mono format); all neural inference is run inside a reproducible Docker CUDA container. In Table 6, numbers without a marker are as reported in the cited papers (different test sets) and rows marked  $^\dagger$

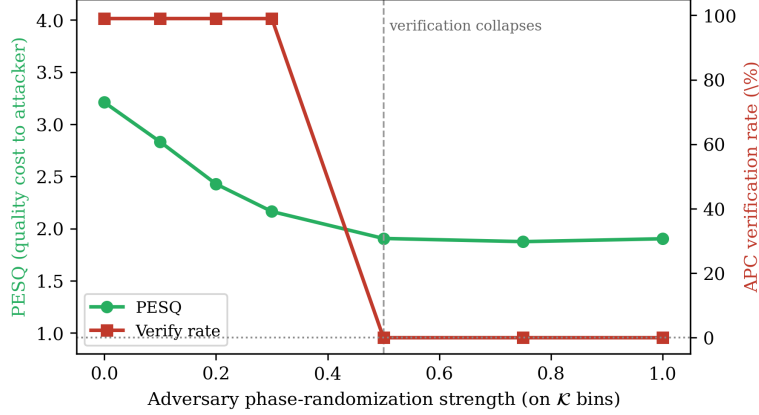


Figure 5: White-box erasure trade-off: verification only collapses at  $\alpha \geq 0.5$ , by which point PESQ has dropped  $\approx 1.3$  points and LSD has  $3.5\times$ -ed.

Table 5: White-box erasure ( $N=100$ ). Table 6: APC vs. neural baselines on the MP3-128 condition.

| $\alpha$ | PESQ | LSD  | Vfy% |
|----------|------|------|------|
| 0.00     | 3.21 | 10.2 | 99   |
| 0.10     | 2.83 | 28.7 | 99   |
| 0.20     | 2.43 | 31.6 | 99   |
| 0.30     | 2.16 | 33.4 | 99   |
| 0.50     | 1.91 | 34.9 | 0    |
| 0.75     | 1.88 | 34.1 | 0    |
| 1.00     | 1.90 | 32.5 | 0    |

| Method                      | Qual.  | MP3 128         | Sig. | T-free |
|-----------------------------|--------|-----------------|------|--------|
| WavMark [16]                | 4.21 P | $\approx 100\%$ | ×    | ×      |
| AudioSeal [15]              | 4.47 P | $\approx 100\%$ | ×    | ×      |
| SilentCipher [17]           | 47 dB  | 100%            | ×    | ×      |
| WavMark <sup>†</sup> [16]   | 4.11 P | 100.0%          | ×    | ×      |
| AudioSeal <sup>†</sup> [15] | 4.44 P | 100.0%          | ×    | ×      |
| APC phase-only              | 3.26 P | 77.0%           | ✓    | ✓      |
| APC hybrid (ours)           | 3.02 P | 97.5%           | ✓    | ✓      |

are our same-test-set reproductions. “P” = PESQ, “dB” = SDR; “Sig.” flags public-key asymmetric signatures; “T-free” flags zero training data and no GPU. Bit-accuracy for the neural rows is over a 16-bit message, not a 64-byte signed payload. We do *not* claim to outperform neural baselines in robustness; we report honestly and analyse the different design axes.

**What APC gives up.** Neural baselines report higher absolute perceptual quality under identity conditions; our same-test-set reproductions confirm the cited numbers within 0.1 P (Table 6, <sup>†</sup> rows), which we take as evidence that the published comparison is fair rather than test-set artefact. WavMark, AudioSeal, and SilentCipher are also typically evaluated at much shorter payloads (16–32 bits), which makes their raw BER numbers easier to stabilise than the 1,160-bit signed payload APC carries per 10 s clip. The 100% bit-accuracy figure for both neural baselines therefore covers a 16-bit symmetric message, not a 64-byte asymmetric signature; carrying an Ed25519 signature on either of these decoders would require either  $\sim 40\times$  more capacity or a chained signing layer such as APC.

**What APC offers in exchange.** (i) *Asymmetric signing*: any user with the decoder model of a neural baseline can also forge its marks; APC embeds a full Ed25519 signature, so only the private-key holder can mint a valid watermark even if the entire extractor is public. (ii) *No training, data, or GPU*: 71.0/57.9 ms embed/extract per 10 s clip on a single CPU thread (hybrid); nothing to retrain when a new codec appears. (iii) *No model drift*: behaviour is deterministic and analysable from  $N_{\text{FFT}}$ ,  $\mathcal{K}$ , and RS parameters alone, which is desirable in certified provenance pipelines. (iv) *Orthogonality*: APC can be stacked *behind* a neural watermark – the neural mark survives codecs, APC adds signing – which we view as the most promising production direction.

#### 4.4 Cost and reproducibility

Single-thread per-clip latency on CPU (10 s 44.1 kHz mono, AMD Threadripper 5990X) is 79.8/7.2 ms embed/extract for the phase-only coder and 71.0/57.9 ms for the hybrid coder, with RS encode and Ed25519 sign/verify each sub-millisecond. Payload capacity is 92 bps after RS overhead, sufficient for the 49-byte signed message. No GPU is required. The full 1,000-clip  $\times$  8-attack hybrid sweep used in this paper completes in 228 s wall-clock with 16 parallel workers on the



same CPU; the 16-attack phase-only and hybrid sweeps used for the appendix figures took 312 s and 448 s respectively, and the 100-clip  $\times$  7- $\alpha$  white-box erasure sweep ran in under two minutes. The supplementary archive ships both coders, the payload manager, the eight-attack benchmark driver, the white-box erasure attack, plotting scripts, the exact `test-clean` path list, seed 42, payload hash, public key, full STFT/QIM parameters, FFmpeg invocations, library versions, and per-sample failure-stage labels in three JSON envelopes.

## 5 Discussion

APC is designed for capture-time provenance pipelines—settings such as C2PA signing, broadcast ingest, or archival capture—where high-quality masters are retained and lower-bitrate derivatives (e.g., MP3/OGG-128, FLAC, or resampled versions) are distributed downstream. Our evaluation matches this regime. In this context, APC provides non-repudiable authentication without requiring access to training data, model checkpoints, or specialized hardware.

Security follows directly from the underlying signature scheme: forging a watermark reduces to breaking Ed25519, and therefore the discrete logarithm problem on Curve25519. As with any perceptual watermark, removal is always possible in principle; the relevant question is the cost of doing so. We quantify this cost in terms of perceptual degradation (PESQ in Table 3, LSD in Fig. 7, App. A). This contrasts with metadata-only signing, where removal is silent: stripped metadata leaves no trace, whereas removing an APC watermark manifests as a failed verification.

Several limitations remain. First, performance degrades on clips with sparse spectral content (Fig. 8, App. A). Second, the current design uses a fixed  $\pm\pi/2$  mapping from bits to phase; adapting the embedding strength to the signal could improve robustness. Third, we evaluate only a simple linear-blend white-box erasure attack (§4.2); stronger optimization-based attacks would likely tighten the trade-off between erasure strength and perceptual quality. Finally, we do not yet study the analog hole (record–replay) or full platform-level transcoding chains.

## 6 Conclusion

We propose Asymmetric Phase Coding (APC), a training-free signal-level watermarking scheme that binds a full Ed25519 digital signature to the audio waveform through a hybrid phase + magnitude-QIM channel under a public-key threat model. On  $N=1,000$  LibriSpeech `test-clean` clips across eight attack configurations—identity, MP3 128 kbps, OGG 128 kbps, FLAC, 16 kHz round-trip resampling, 8 kHz low-pass, and 10%/20% end-cropping—the hybrid coder achieves cryptographic verification rates between 97.5% and 98.3% at mean PESQ = 3.02 and tens-of-milliseconds CPU latency, with the magnitude channel rescuing 14.1% of MP3-128 and 9.2% of OGG-128 clips for which the phase channel alone fails. A white-box erasure analysis lower-bounds the perceptual cost of removing the watermark: verification only collapses at  $\alpha \geq 0.5$  phase randomization, by which point mean PESQ has dropped  $\approx 1.3$  points and LSD has grown  $\sim 3.5\times$ , in contrast to the zero-cost stripping of a metadata-only signature. APC therefore occupies a different point in the audio-watermarking design space than recent neural baselines: it gives up roughly 1.2–1.5 PESQ relative to AudioSeal/WavMark/SilentCipher in exchange for asymmetric public-key signing, zero training data or GPU, deterministic and analysable behaviour, and a clean stacking path behind a neural watermark for C2PA-style provenance pipelines. We view this last composition—neural mark for codec survival, APC for non-repudiable signing—as the most promising direction for production deployment.

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## A Supplementary figures

This appendix collects four supporting visualisations whose underlying numbers are already reported in the main paper. They are included for readers who prefer a per-attack visual rendering, but are not required to follow the main argument: Fig. 6 re-displays the cryptographic verification rate (Verify% column of Table 3); Fig. 7 plots LSD across the same eight attacks (numerically referenced in §4.2 and §5); Fig. 8 is the per-clip BER heatmap that motivates the content-adaptive limitation discussed in §5; Fig. 9 renders the NC survivability of Table 3 as a radar plot.

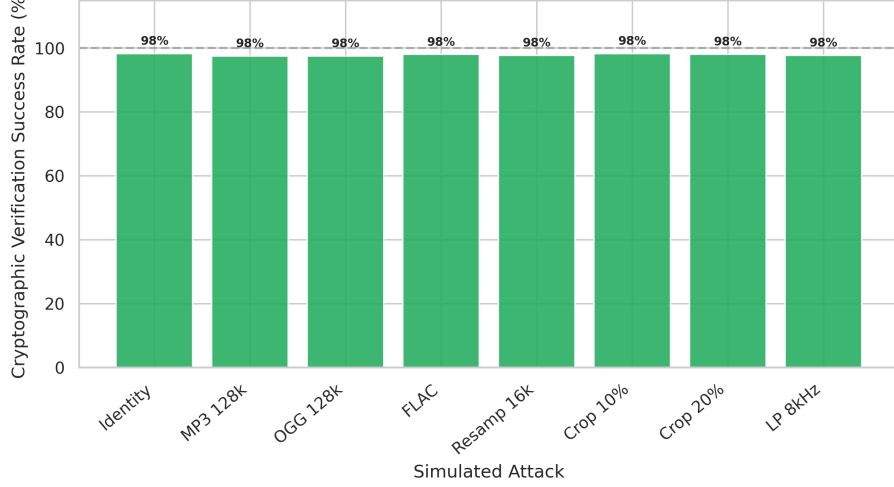


Figure 6: Cryptographic verification rate per attack. Hybrid APC verifies between 97.5% and 98.3% on every condition; the same numbers are tabulated in the Verify% column of Table 3.

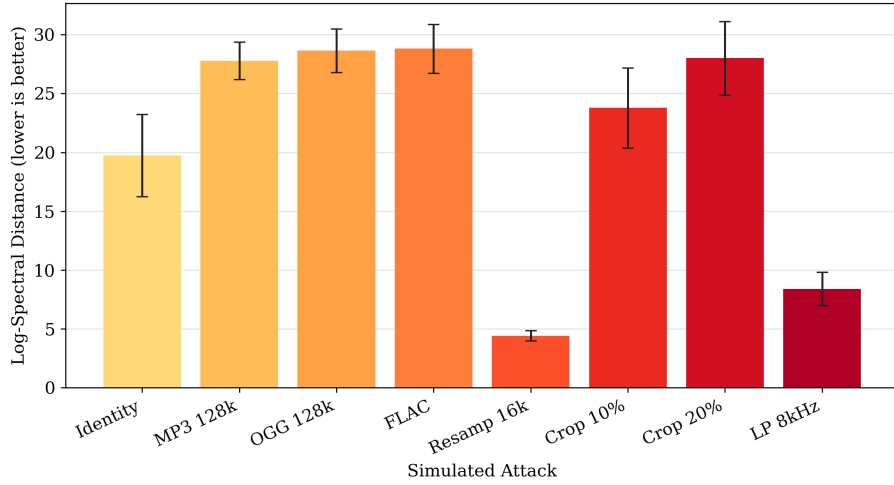


Figure 7: Log-spectral distance (LSD, lower is better) across the eight attacks. Resampling-16k preserves the spectrum almost exactly; codecs and low-pass produce the largest LSD shifts. The numerically corresponding values appear in Table 5 (under-attack rows) and are discussed in §4.2, §5.

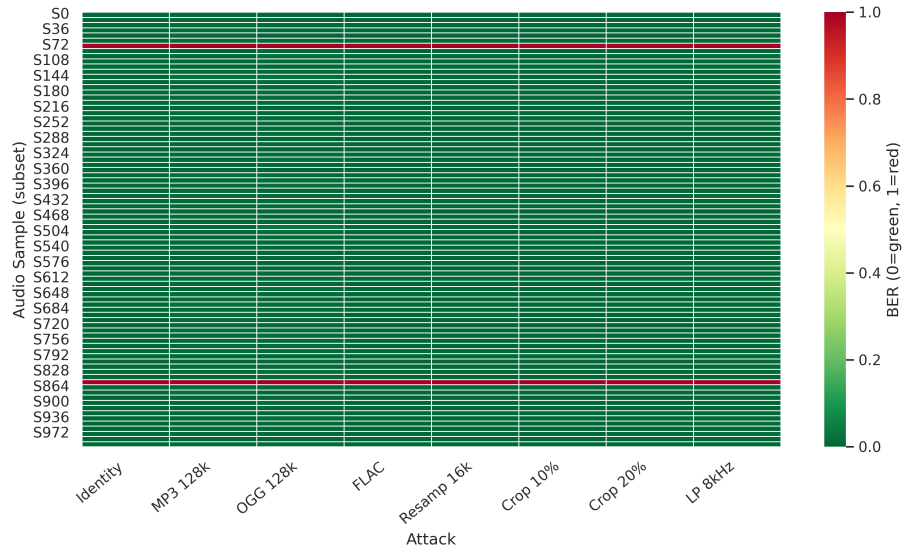


Figure 8: Per-sample BER heatmap (subset of LibriSpeech test-clean clips, x-axis = attack, y-axis = audio sample id). Most cells are bright green ( $\text{BER} \approx 0$ ); the residual failure mass is content-dependent rather than attack-dependent, motivating the content-adaptive embedding strength discussed in §5.

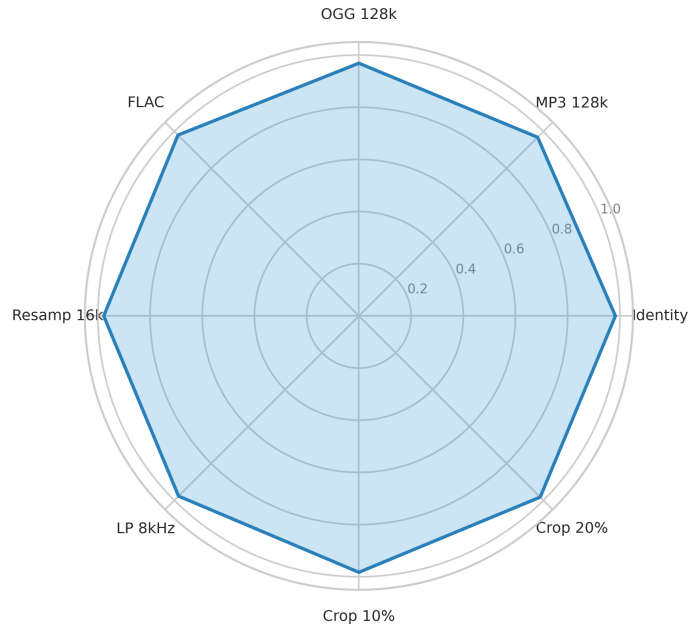


Figure 9: NC survivability radar across key codec/channel attacks. Every axis is at or above 0.96; this is the visual rendering of the NC column of Table 3.